

Universitat Politècnica de Catalunya
Grau en Enginyeria Física · Mètodes matemàtics 1

Annex: Codi MATLAB

Estudi dinàmic del Model FitzHugh-Nagumo

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El codi següent implementa la simulació numèrica del Model de FitzHugh-Nagumo mitjançant MATLAB. Per a cada valor del corrent extern $I_{\text{ext}} \in \{0, I_c, 0.7\}$, es calcula el punt d'equilibri via el mètode de Newton, se'n classifica l'estabilitat a partir de la traça i el determinant de la matriu jacobiana, i es resolen numèricament les equacions diferencials amb ode45 per representar el retrat de fases i l'evolució temporal del voltatge.

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1 %% =====
2 % Model de FitzHugh-Nagumo
3 % Parametres: a=0.7, b=0.8, tau=12.5
4 % =====
5 clear all; close all; clc; format long
6
7 set(groot, 'defaultFigureColor', 'w') % fons blanc a totes les figures
8
9 %% --- Parametres del model -----
10 a = 0.7;
11 b = 0.8;
12 tau = 12.5;
13
14 %% --- Corrent critic analitic (bifurcacio de Hopf) -----
15 % La bifurcaci es produeix a la branca NEGATIVA de la c bica
16 % (equilibri fisiol gicament rellevant, V* < 0 per a I petit)
17 Vc = -sqrt(1 - b/tau); % branca negativa
18 Wc = (Vc + a) / b;
19 Ic = Wc - Vc + (Vc^3)/3;
20
21 fprintf('=== Bifurcacio de Hopf ===\n')
22 fprintf('V*_c = %.6f\n', Vc)
23 fprintf('W*_c = %.6f\n', Wc)
24 fprintf('I_c = %.6f\n\n', Ic)
25
26 %% --- Valors de I_ext a estudiar -----
27 I_vals = [0, round(Ic, 4), 0.7];
28 etiques = {'Rep s (Focus atractor)', ...
29           'Bifurcaci de Hopf (I_c \approx %.4f)', ...
30           'R gim oscil latori (Node repulsor)'};
31
32 %% --- Opcions d'integracio -----
33 prec = odeset('RelTol', 1e-8, 'AbsTol', 1e-8);
34 v = linspace(-2.5, 2.5, 600);
35
36 %% --- Colors -----
37 col_nulcV = [0.85 0.15 0.15];
38 col_nulcW = [0.00 0.55 0.75];
39 col_traj = [0.50 0.00 0.50];

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40 col_eq      = [0.10 0.70 0.10];
41
42 %% === FIGURES 1 3 : retrat de fases + dinamica temporal =====
43 X0_list = [2 2; -2 0.5; 0 -0.3; 1.5 -0.2; -1 2];
44
45 for i = 1:3
46     Iext = I_vals(i);
47
48     FHN    = @(t,X) [X(1) - (X(1)^3)/3 - X(2) + Iext;
49                     (X(1) + a - b*X(2)) / tau];
50     nulcV = @(v)  v - (v.^3)/3 + Iext;
51     nulcW = @(v)  (v + a) / b;
52
53     % Punt d'equilibri (Newton, punt inicial proper a -1)
54     fundif = @(v) (v.^3)/3 - v*(1 - 1/b) + a/b - Iext;
55     [xk, ~, ~] = newtonDnD(-1, 1e-10, 50, fundif, '0');
56     vx = xk(end);
57     wx = nulcW(vx);
58
59     % Classificacio
60     J      = [1 - vx^2, -1; 1/tau, -b/tau];
61     Tj     = trace(J);
62     Dj     = det(J);
63     Deltj  = Tj^2 - 4*Dj;
64
65     if      Dj < -1e-10, c = 'Sella';
66     elseif Dj >  1e-10
67         if Deltj < -1e-10
68             if      Tj < -1e-10, c = 'Focus atractor';
69             elseif Tj >  1e-10, c = 'Focus repulsor';
70             else,              c = 'Centre'; end
71         else
72             if Tj < -1e-10, c = 'Node atractor';
73             else,          c = 'Node repulsor'; end
74         end
75     else, c = 'Degenerat'; end
76
77     fprintf('I_ext = %.4f  ->  Equilibri (%.4f, %.4f)  |  %s\n', Iext,
78     vx, wx, c)
79     fprintf('      T = %.6f      D = %.6f      Delta = %.6f\n\n', Tj, Dj, Deltj)
80
81     % Camp vectorial normalitzat
82     [Vg, Wg] = meshgrid(linspace(-2.5,2.5,18), linspace(-0.4,2.4,13));
83     dVg = Vg - (Vg.^3)/3 - Wg + Iext;
84     dWg = (Vg + a - b*Wg) / tau;
85     nrm = sqrt(dVg.^2 + dWg.^2) + 1e-12;
86     dVn = dVg ./ nrm;
87     dWn = dWg ./ nrm;
88
89     fig = figure(i);
90     set(fig, 'Units','centimeters', 'Position',[2+3*i 2 22 8])
91
92     % Retrat de fases
93     subplot(1,2,1); hold on; grid on; box on
94     quiver(Vg, Wg, dVn, dWn, 0.45, 'Color',[0.78 0.78 0.78], ...
95           'LineWidth',0.6, 'MaxHeadSize',0.4)
96     plot(v, nulcV(v), '-','Color',col_nulcV, 'LineWidth',1.8)
97     plot(v, nulcW(v), '-','Color',col_nulcW, 'LineWidth',1.8)

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97     for k = 1:size(X0_list,1)
98         X0 = X0_list(k,:);
99         [~, Xout] = ode45(FHN, [0 120], X0, prec);
100         plot(Xout(:,1), Xout(:,2), '-', 'Color',col_traj, ...
101             'LineWidth',1.0, 'HandleVisibility','off')
102         plot(X0(1), X0(2), '.', 'Color',col_traj, 'MarkerSize',8, ...
103             'HandleVisibility','off')
104     end
105     plot(vx, wx, '*', 'Color',col_eq, 'MarkerSize',10, 'LineWidth',1.8)
106     xlabel('$V$', 'Interpreter','latex', 'FontSize',12)
107     ylabel('$W$', 'Interpreter','latex', 'FontSize',12)
108     title(sprintf('(%s)', c), 'FontSize',11)
109     legend({'Nulclina $\dot{V}=0$', 'Nulclina $\dot{W}=0$', 'Equilibri'},
110         ...
111         'Interpreter','latex', 'FontSize',9, 'Location','northeast')
112     axis([-2.5 2.5 -0.4 2.4])
113     set(gca,'FontSize',10)
114
115     % Dinamica temporal
116     subplot(1,2,2); hold on; grid on; box on
117     [Tout, Xout] = ode45(FHN, [0 120], X0_list(1,:), prec);
118     plot(Tout, Xout(:,1), '-', 'Color',col_traj, 'LineWidth',1.6)
119     ylabel(vx, '--', 'Color',col_eq, 'LineWidth',1.0, ...
120         'Label','$V^*$', 'Interpreter','latex', ...
121         'LabelHorizontalAlignment','left', 'FontSize',9)
122     xlabel('Temps', 'FontSize',12)
123     ylabel('$V(t)$', 'Interpreter','latex', 'FontSize',12)
124     title(sprintf('$I_{\mathrm{ext}} = %.4f$', Iext), ...
125         'Interpreter','latex', 'FontSize',11)
126     set(gca,'FontSize',10)
127
128     sgtitle(etiques{i}, 'FontSize',12, 'FontWeight','bold')
129 end
130 %% == FIGURA 4: nulclines per a diversos I_ext ==
131 figure(4)
132 set(gcf,'Units','centimeters','Position',[2 2 16 10])
133 hold on; grid on; box on
134
135 I_sweep = 0 : 0.3 : 1.5;
136 cmap = parula(length(I_sweep));
137
138 for k = 1:length(I_sweep)
139     Iext = I_sweep(k);
140     nulcV = @(v) v - (v.^3)/3 + Iext;
141     plot(v, nulcV(v), '-', 'Color',cmap(k,:), 'LineWidth',1.4)
142 end
143
144 nulcW = @(v) (v + a)/b;
145 plot(v, nulcW(v), 'k--', 'LineWidth',1.8)
146
147 nulcVc = @(v) v - (v.^3)/3 + Ic;
148 plot(v, nulcVc(v), 'r-', 'LineWidth',2.2)
149 plot(Vc, Wc, 'r*', 'MarkerSize',10, 'LineWidth',1.8)
150
151 xlabel('$V$', 'Interpreter','latex', 'FontSize',12)
152 ylabel('$W$', 'Interpreter','latex', 'FontSize',12)
153 title('Nulclines en funci de $I_{\mathrm{ext}}$', ...

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154     'Interpreter','latex', 'FontSize',12)
155
156 legEntries = arrayfun(@(I) sprintf('$I_{\mathrm{ext}}$=%.1f$',I), ...
157     I_sweep, 'UniformOutput',false);
158 legEntries{end+1} = sprintf('$I_c \approx %.4f$ (bifurcaci )', Ic);
159 legEntries{end+1} = 'Nulclina $\dot{W}=0$';
160 h1 = plot(nan,nan,'r-','LineWidth',2);
161 h2 = plot(nan,nan,'k--','LineWidth',1.8);
162 hc = gobjects(length(I_sweep),1);
163 for k = 1:length(I_sweep)
164     hc(k) = plot(nan,nan,'-','Color',cmap(k,:), 'LineWidth',1.4);
165 end
166 legend([hc; h1; h2], legEntries, 'Interpreter','latex', ...
167     'FontSize',8, 'Location','northwest')
168 axis([-2.5 2.5 -1 4])
169 set(gca,'FontSize',10)
170
171 %% == FIGURA 5: Diagrama de bifurcacio ==
172 figure(5)
173 set(gcf,'Units','centimeters','Position',[2 2 16 10])
174 hold on; grid on; box on
175
176 I_sweep2 = linspace(0, 1.0, 120);
177 V_max = zeros(size(I_sweep2));
178 V_min = zeros(size(I_sweep2));
179 V_eq = zeros(size(I_sweep2));
180
181 for k = 1:length(I_sweep2)
182     Iext = I_sweep2(k);
183     FHN2 = @(t,X) [X(1)-(X(1)^3)/3-X(2)+Iext; (X(1)+a-b*X(2))/tau];
184
185     [Tout2, Xout2] = ode45(FHN2, [0 400], [0.5; 0.5], prec);
186
187     idx = Tout2 > 0.75*Tout2(end);
188     V_ss = Xout2(idx, 1);
189
190     V_max(k) = max(V_ss);
191     V_min(k) = min(V_ss);
192
193     % Newton amb punt inicial -1 (branca correcta)
194     fundif2 = @(v) (v.^3)/3 - v*(1-1/b) + a/b - Iext;
195     [xk2,~,~] = newtonDnD(-1, 1e-10, 50, fundif2, '0');
196     V_eq(k) = xk2(end);
197 end
198
199 col_cicle = [0.20 0.45 0.75];
200 col_eq_st = [0.00 0.00 0.00];
201 col_eq_unst = [0.50 0.50 0.50];
202 col_ic = [0.85 0.40 0.05];
203
204 fill([I_sweep2, fliplr(I_sweep2)], [V_max, fliplr(V_min)], ...
205     col_cicle, 'FaceAlpha',0.20, 'EdgeColor','none', ...
206     'DisplayName','Amplitud cicle 1 mit')
207 plot(I_sweep2, V_max, '-', 'Color',col_cicle, 'LineWidth',2.0, ...
208     'DisplayName','$V_{\max}$ / $V_{\min}$')
209 plot(I_sweep2, V_min, '-', 'Color',col_cicle, 'LineWidth',2.0, ...
210     'HandleVisibility','off')
211 plot(I_sweep2(I_sweep2 <= Ic), V_eq(I_sweep2 <= Ic), '-', ...

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212     'Color',col_eq_st, 'LineWidth',2.0, 'DisplayName','Equilibri
    estable')
213 plot(I_sweep2(I_sweep2 >= Ic), V_eq(I_sweep2 >= Ic), '--', ...
214     'Color',col_eq_unst, 'LineWidth',2.0, 'DisplayName','Equilibri
    inestable')
215
216 % Ic: sense HandleVisibility per evitar 'data1' a la llegenda
217 xline(Ic, '--', 'Color',col_ic, 'LineWidth',2.0, ...
218     'HandleVisibility','off', ...
219     'Label',sprintf('$I_c \approx %.4f$', Ic), ...
220     'Interpreter','latex', 'FontSize',10, ...
221     'LabelVerticalAlignment','bottom', ...
222     'LabelHorizontalAlignment','right')
223
224 % Entrada manual d'Ic a la llegenda
225 plot(nan, nan, '--', 'Color',col_ic, 'LineWidth',2.0, ...
226     'DisplayName',sprintf('$I_c \approx %.4f$', Ic))
227
228 xlabel('$I_{\mathrm{ext}}$', 'Interpreter','latex', 'FontSize',13)
229 ylabel('$V^*$', 'Interpreter','latex', 'FontSize',13)
230 title('Diagrama de bifurcacions --- Model de FitzHugh-Nagumo', '
    FontSize',12)
231 legend('Interpreter','latex', 'FontSize',10, 'Location','northwest')
232 axis([0 1.0 -2.2 2.2])
233 set(gca,'FontSize',10)
234
235 %% === Funcio de Newton ===
236 function [xk, res, it] = newtonDnD(x0, tol, itmax, fun, dfun)
237     xk = x0;
238     fk = fun(xk);
239     res = abs(fk);
240     it = 0;
241     dx = 1e-7;
242     while it < itmax && res(end) > tol
243         if ischar(dfun) && strcmp(dfun,'0')
244             dfk = (fun(xk(end) + dx) - fk) / dx;
245         else
246             dfk = dfun(xk(end));
247         end
248         xk = [xk, xk(end) - fk/dfk];
249         fk = fun(xk(end));
250         res = [res, abs(fk)];
251         it = it + 1;
252     end
253 end
254
255 % -----
256 %% Funcio f(V)
257 % -----
258
259 v = linspace(-3, 3, 600);
260
261 for b = [0.1, 3]
262
263     f = @(v) -v.^2 + 1 - 1/b;
264
265     figure
266     hold on; grid on; box on

```

```

267
268 % Funcio
269 plot(v, f(v), 'b-', 'LineWidth', 2, 'DisplayName', sprintf("$f'(V)$", b))
270
271 % Eix horitzontal
272 yline(0, 'k--', 'LineWidth', 1.2, 'DisplayName', '$f'(V)=0$')
273
274 % Arrels (noms si b > 1)
275 if b > 1
276     arrels = sqrt(1 - 1/b);
277     fprintf('Per b = %.1f, arrels a V = %.4f\n', b, arrels)
278     plot([ arrels, -arrels], [0, 0], 'ro', ...
279         'MarkerSize', 8, 'LineWidth', 1.5, ...
280         'DisplayName', sprintf('$V_{\pm} = \pm %.4f$', arrels))
281 else
282     fprintf('Per b = %.1f, no hi ha arrels reals\n', b)
283 end
284
285 % Format
286 xlabel('$V$', 'Interpreter', 'latex', 'FontSize', 13)
287 ylabel("$f'(V)$", 'Interpreter', 'latex', 'FontSize', 13)
288 title(sprintf("$f'(V) = -V^2 + 1 - 1/b$, \quad b = %.1f$", b), ...
289     'Interpreter', 'latex', 'FontSize', 12)
290 legend('Interpreter', 'latex', 'FontSize', 10, 'Location', 'north')
291 axis([-2.5 2.5 -2 1.5])
292 set(gca, 'FontSize', 10)
293
294 end

```

Listing 1: Simulació numèrica del Model de FitzHugh-Nagumo.